IBM AML – Transaction and Fraud Analysis Dashboard Using Power BI

REPORT

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1. Introduction

Power BI is a business intelligence and data visualization tool used to analyse data and create interactive dashboards. This project focuses on analysing banking transactions, account details and fraud alerts using Power BI.

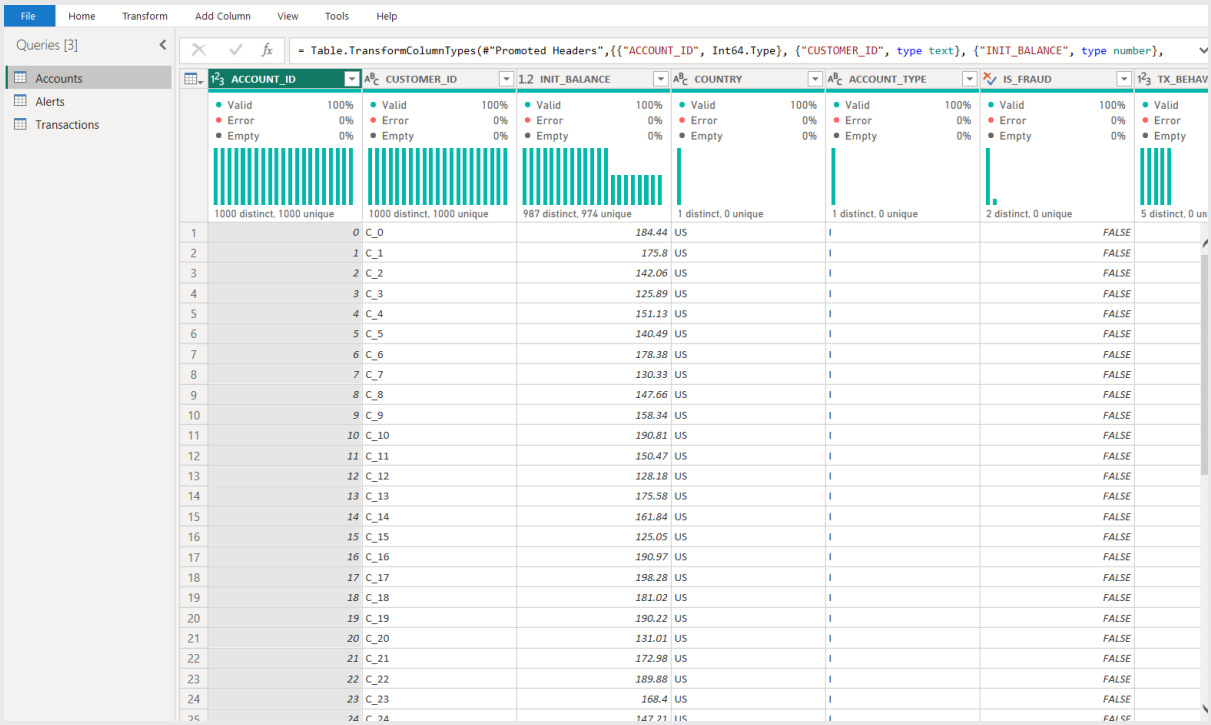
The main objective of this activity is to identify fraudulent transactions, understand transaction patterns, analyse account balances and help management make better financial decisions.

2. Steps Involved

Step 1: Importing the Dataset

- Open Power BI Desktop.
- Import the three CSV files: Accounts, Transactions and Alerts.
- Save the Power BI file as IBM AML.pbix.
- Check the imported data in Data View and Model View.
- Identify identifier, category, numerical and date fields.

Step 2: Data Inspection and Cleaning



- Open Power Query Editor.
- Rename the tables by capitalizing their first letters.

- Check Accounts[ACCOUNT_ID] for null or blank values.
- Keep Transactions[ALERT_ID] values of -1 without replacing them.
- Check duplicate TX_ID values in the Transactions table.
- Allow repeated ALERT_ID values in the Alerts table.
- Investigate unusually large or negative transaction amounts and initial balances.
- Ensure IS_FRAUD has a True/False data type.
- Keep TIMESTAMP as a numeric field.

Step 3: Creating Relationships

Create the following relationships using Manage Relationships:

From	To	Relationship
Accounts[ACCOUNT_ID]	Transactions[SENDER_ACCOUNT_ID]	One-to-many (Active)
Accounts[ACCOUNT_ID]	Transactions[RECEIVER_ACCOUNT_ID]	One-to-many (Inactive)
Transactions[TX_ID]	Alerts[TX_ID]	One-to-many (Active)

Set the cross-filter direction to Single for all three relationships.

Step 4: Creating DAX Measures

DAX measures are used to calculate important values from the dataset.

Create the following six measures:

1. Total Transaction Amount

Total Transaction Amount =

SUM(Transactions[TX_AMOUNT])

2. Total Transactions

Total Transactions =

COUNT(Transactions[TX_ID])

3. Total Accounts

Total Accounts =

DISTINCTCOUNT(Accounts[ACCOUNT_ID])

4. Average Transaction Amount

Average Transaction Amount =

AVERAGE(Transactions[TX_AMOUNT])

5. Total Initial Balance

Total Initial Balance =

SUM(Accounts[INIT_BALANCE])

6. Fraudulent Transactions

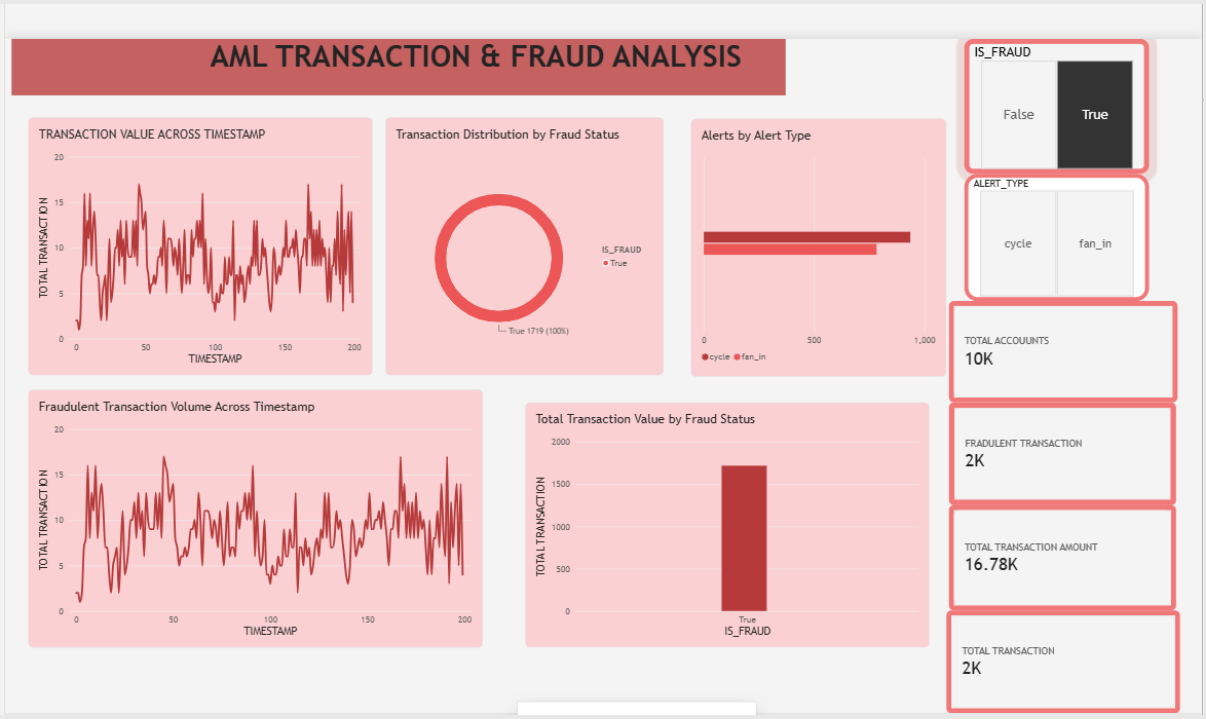
Fraudulent Transactions =

```
CALCULATE(
   COUNT(Transactions[TX_ID]),
   Transactions[IS_FRAUD] = TRUE()
)
```

After creating the measures, format them according to their values. Use Decimal Number for transaction amounts and Whole Number for transaction and account counts.

Step 5: Creating the Dashboard

Create the following visualizations in the report view.



1. Card Visuals

Create four cards to display:

- Total Transactions
- Total Transaction Amount
- Total Accounts
- Fraudulent Transactions

2. Line Chart – Transaction Volume Across Timestamp

- X-axis: TIMESTAMP
- Y-axis: Total Transactions

This chart displays transaction volume across different timestamps.

3. Donut Chart – Transaction Distribution by Fraud Status

- Legend: IS_FRAUD
- Values: Total Transactions

This chart shows the distribution of fraudulent and non-fraudulent transactions.

4. Line Chart – Fraudulent Transaction Volume Across Timestamp

- X-axis: TIMESTAMP
- Y-axis: Total Transactions

- Apply a visual-level filter to IS_FRAUD and select True.

This chart helps identify fraudulent transaction patterns over time.

5. Clustered Bar Chart – Alerts by Alert Type

- Y-axis: ALERT_TYPE
- X-axis: ALERT_ID (Count)

This chart displays the number of alerts under different alert categories.

6. Clustered Column Chart – Total Transaction Value by Fraud Status

- X-axis: IS_FRAUD
- Y-axis: Total Transaction Amount

This chart compares transaction values between fraudulent and non-fraudulent transactions.

Step 6: Creating an Interactive Dashboard

- Duplicate the existing AML Transaction & Fraud Analysis page.
- Rename the duplicated page as Interactive AML Dashboard.
- Keep the original dashboard unchanged.
- Rearrange the visualizations to create a clear dashboard layout.

Step 7: Adding Slicers

Add two slicers to the Interactive AML Dashboard.

1. Fraud Status

- Field: Transactions[IS_FRAUD]
- Set the slicer title as Fraud Status.

2. Alert Type

- Field: Alerts[ALERT_TYPE]
- Set the slicer title as Alert Type.

These slicers allow users to filter the dashboard based on fraud status and alert categories.

Step 8: Checking Visual Interactions

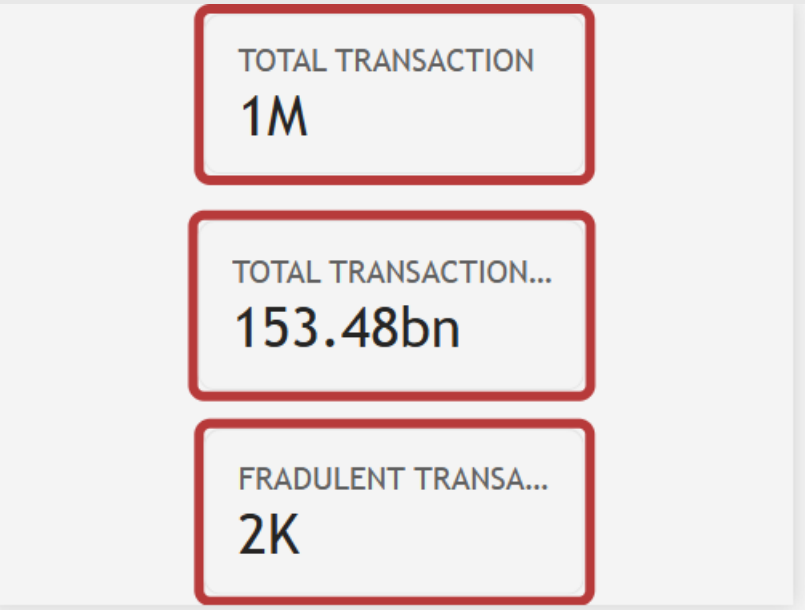
- Select the Fraud Status slicer.
- Open the Edit Interactions option.

- Check the filter and none options for the different visuals.
- Ensure that the slicer affects only the required visuals.

Step 9: Creating a Tooltip Page

- Create a new report page.
- Rename it as AML Tooltip.
- Change the page type to Tooltip.
- Turn on Allow use as tooltip.
- Add three Card visuals:
    1. Total Transactions
    2. Total Transaction Amount
    3. Fraudulent Transactions
- Return to the Interactive AML Dashboard.
- Select the Transaction Volume Across Timestamp line chart.
- Enable Tooltips and select Report Page.
- Choose AML Tooltip as the tooltip page.

This allows users to view additional transaction information when hovering over the chart.



Step 10: Applying a Power BI Theme

- Open the View ribbon.

- Select the Themes dropdown.
- Choose a suitable built-in theme.
- Observe the changes in chart colours, text and background.
- Select a theme that makes the dashboard attractive and easy to understand.

3. Conclusion

This activity helped me understand how to import, inspect and transform banking datasets using Power BI. I learned to create relationships, write DAX measures and design interactive dashboards using different visualizations. The dashboard helps analyse transaction amounts, account details, fraud status and alert types. It also helps management identify suspicious activities, monitor fraudulent transactions and make better financial decisions.